

PYTHON DATA VISUALISATION SERIES

Cheat Sheet Report

Maps, Waffles, WordClouds & Seaborn

A Practical Reference for Advanced
Data Visualisation in Python

Data Visualization with Python

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Abstract

This report summarises the key functions and visualisations from the cheat sheet on advanced data visualisation tools in Python. The libraries covered include **Folium** for interactive maps, **PyWaffle** for waffle charts, **WordCloud** for text visualisation, and **Seaborn** for statistical plots. Each section provides a brief description, illustrative code, and example output.

1 Introduction

The cheat sheet provides a quick reference for creating a variety of specialised plots using Python libraries. It covers:

- **Folium** — interactive maps with markers, circles, and choropleth layers.
- **PyWaffle** — waffle charts for categorical data.
- **WordCloud** — word clouds to highlight text frequencies.
- **Seaborn** — statistical bar, count, and regression plots.

The following sections illustrate these tools with example visualisations and short, runnable code snippets.

2 Folium Maps

Folium builds on the *Leaflet.js* library to produce interactive maps directly from Python. Key functions are illustrated below.

2.1 Basic Map

Create a map with a given centre and zoom level:

```
folium.Map(location=[56.130, -106.35], zoom_start=4)
```

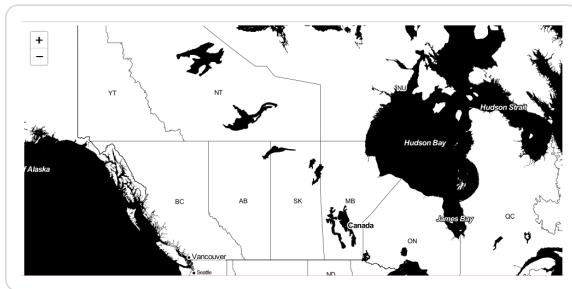


Figure 1: Basic world map with Canada centred.

2.2 Markers with Different Tiles

Add markers with custom tiles (Stamen Toner and Stamen Terrain):

```
folium.Marker(location=[56.130, -106.35],  
              tooltip='Marker',  
              tiles='Stamen Toner').add_to(map)
```



(a) Stamen Toner tile.



(b) Stamen Terrain tile.

Figure 2: Marker examples with different tile styles.

2.3 Circle Marker

Add a circle with a specified radius and fill opacity:

```
folium.features.CircleMarker(location=[56.130, -106.35],  
                             radius=1000, color='red',  
                             fill_opacity=0.5).add_to(map)
```

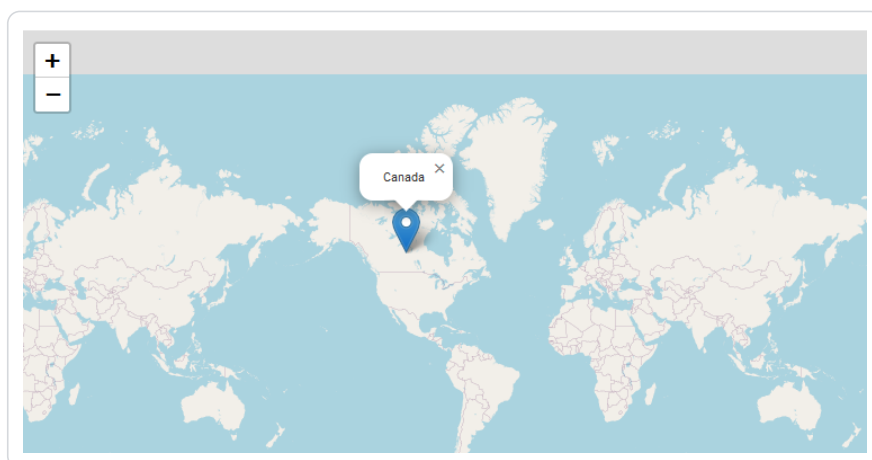


Figure 3: Circle marker on the map.

2.4 Choropleth Map

Create a choropleth map using a GeoJSON file and a data table:

```
folium.Choropleth(geo_data=world_geo, data=df_can,
                  columns=['Country', 'Total'],
                  key_on='feature.properties.name',
                  fill_color='YlOrRd').add_to(map)
```

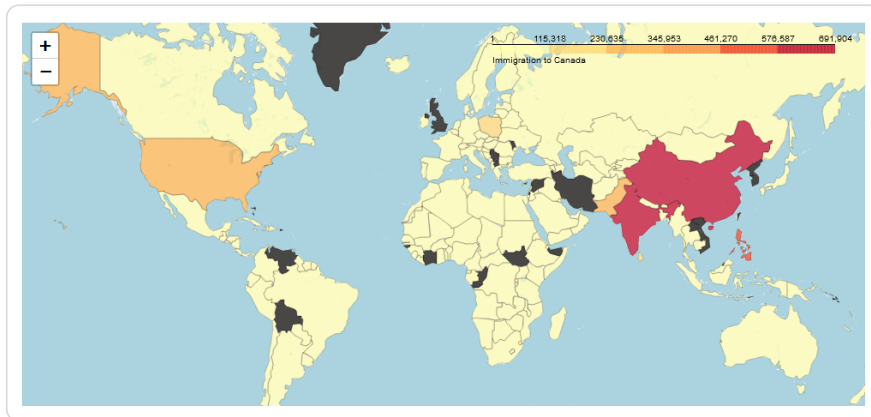


Figure 4: Choropleth map showing immigration to Canada.

3 PyWaffle

PyWaffle creates waffle charts to represent proportions using a grid of squares, offering an intuitive alternative to pie charts.

3.1 Waffle Chart

Create a waffle chart with specified rows, columns, and values:

```
plt.figure(FigureClass=Waffle, rows=20, columns=30,
          values=df_dsn['Total'], cmap_name='tab20')
```

3.2 Legend, Title, and Labels

Add a legend, title, and labels to the waffle chart using:

```
waffle_chart.legend(loc='upper left', bbox_to_anchor=(1, 1))
waffle_chart.set_title('Waffle Chart Title')
waffle_chart.set_labels(['Label 1', 'Label 2', ...])
```

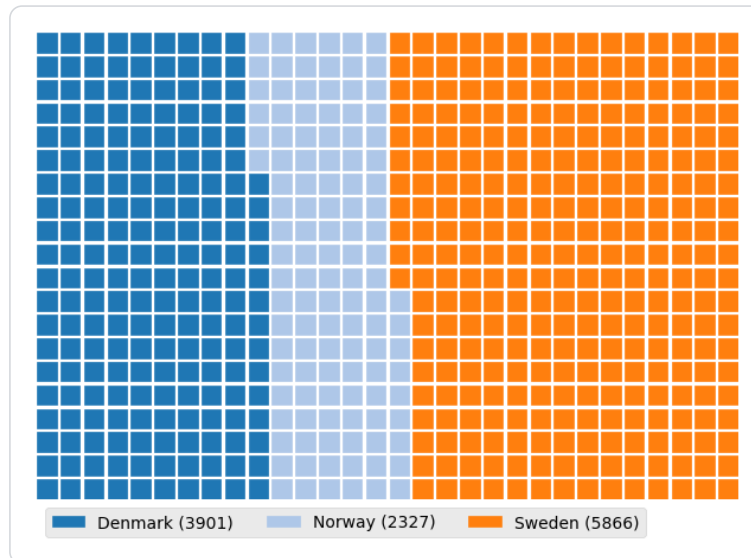


Figure 5: Waffle chart showing categorical distribution.

4 WordCloud

The WordCloud library generates word clouds from text data, with many customisation options such as masks, colour maps, and stop-word filtering.

```
wordcloud = WordCloud(background_color='white',
                       max_words=2000,
                       mask=alice_mask,
                       stopwords=stopwords).generate(text)
plt.imshow(wordcloud, interpolation='bilinear')
```

5 Seaborn

Seaborn provides high-level interfaces for drawing attractive and informative statistical graphics.

5.1 Bar Plot

Visualise the relationship between a categorical and a numeric variable:

```
sns.barplot(x='Continent', y='Total', data=df_can1)
```

5.2 Count Plot

Display the frequency of each category:

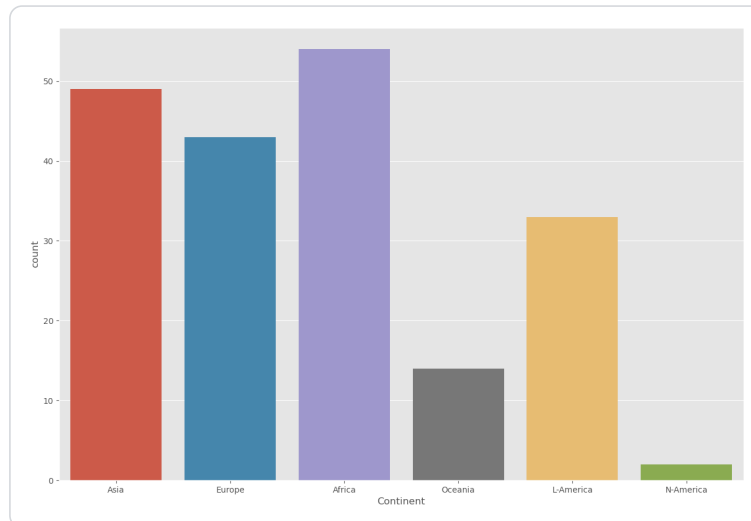


Figure 7: Count plot showing continent frequencies.

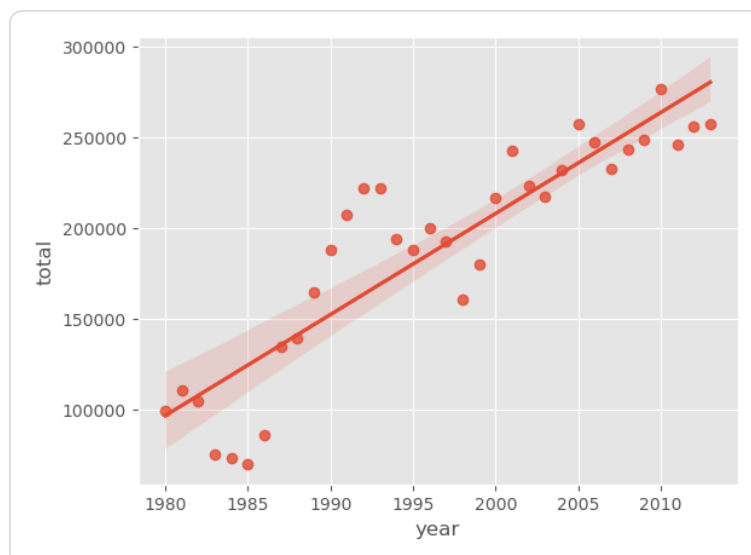


Figure 8: Regression plot showing trend over years.

- **WordCloud** highlights term frequency in text corpora, ideal for exploratory text analysis.
- **Seaborn** streamlines statistical plotting with sensible defaults built on top of Matplotlib.